

Unsupervised Learning - Lab 03 - Basic Clustering

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Clustering Basics – Lab Session Agenda

- Motivation:
 - This third lab session aims to explore the importance of input preparation (e.g., normalization/scaling) and feature selection to obtain a "good" clustering solution.
 - The effectiveness of the clustering is also evaluated.
- Steps:
 - Random feature selection and dataset visualization (Part I - TASK 1,2)
 - Compute proximity matrix, intra-/inter-cluster distances (Part I - TASK 3-6)
 - Proper feature selection and repeat analysis (Part II - TASK 7,8)
 - Scaling and its effect on clustering (Part III - TASK 9,10*)

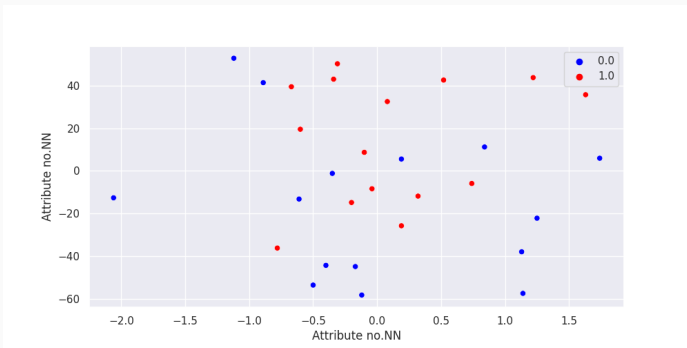
***Optional tasks if time allows**

- Explore input preparation techniques (feature selection, scaling, normalization)
- Evaluate clustering effectiveness based on different metrics
- Hands-on practice with Python libraries: numpy, pandas, scipy, matplotlib, seaborn, sklearn

Dataset Visualization (TASK 1-2)

- The dataset (X) is generated with N objects, M attributes, K=2 groups (classes)
- Load dataset and select two features (feat1, feat2)
- Plot objects using scatterplot (sns.scatterplot)

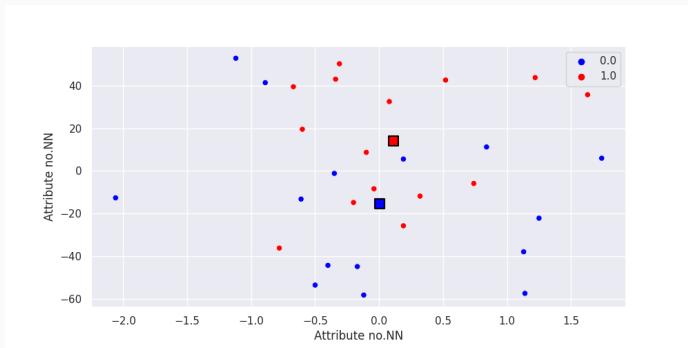
Example Visualization:



Proximity Matrix and Centroids (TASK 3-4)

- Compute centroids as mean point of each group
- Compute and visualize proximity matrix using SciPy's `pdist()`
- Use consistent markers: blue/red for groups, dots for objects, squares for centroids

Example Centroids Visualization:



Intra- and Inter-cluster Distances (TASK 5-6)

- Intra-cluster distance: average pairwise distance within clusters
- Inter-cluster distance: distance between centroids
- Print computed values and analyze clustering effectiveness

Key Question: Is this a good clustering solution?

Feature Selection and Repeat Analysis (TASK 7-8)

- Select new features (feat3, feat4)
- Compute new centroids, proximity matrix, intra-/inter-cluster distances
- Visualize results and compare with previous selection
- Print the following information:
 - your new choice of the features (feat3 and feat4, this time) and a motivation
 - the coordinates of the new centroids
 - the intra- and inter-cluster distances

Scaling and Its Effect on Clustering (TASK 9-10*)

- Apply different scaling transformations:
 - StandardScaler
 - Normalize
 - RobustScaler
 - MinMaxScaler
- Compute new centroids, proximity matrix, intra-/inter-cluster distances
- Compare results before and after scaling
- Print/Produce the following information:
 - the coordinates of the new centroids
 - the two plots to compare the results with/without transformation
 - the new proximity matrix
 - the new intra- and inter-cluster distances
 - repeat the analysis above with another transformation choice